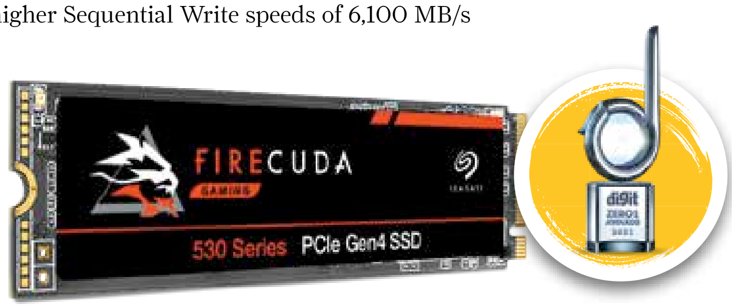
NVMe SSDs

Blitzkrieg

It's been two years since PCIe Gen 4.0 made its way onto consumer boards and led to a flurry of NVMe SSD launches to take advantage of the higher speeds. However, PCIe Gen 4.0 integrated chipsets used to run a little hot and so did the first gen SSDs. If you were getting a PCIe Gen 4.0 SSD, then you'd have to make provision to fit a beefy SSD heat-sink and pick a motherboard that wouldn't cause your graphics card to sit atop the NVMe slot. Those troubles are now gone. Both AMD and Intel platforms now support PCIe Gen 4.0 and SSD manufacturers have been toiling away towards building faster, smarter and power efficient controllers. The Phison E18 controller that came out this year has resulted in several interesting designs that have surprised us with their performance. The Zero1 Awards are all about performance and this year's winner in the NVMe category is...

Winner: Seagate FireCuda 530 NVMe SSD

Being the latest SSD to incorporate the Phison E18 controller, the Seagate FireCuda 530 has managed to get the most out of the E18's capabilities. It's coupled with Micron's 176-layer TLC NAND and is rated for an endurance of 1275 TBW over its life. That's easily twice as much as what most of the competition offers. Only MSI with their Spatium M480 offers similar endurance. On the performance front, the FireCuda 530 managed to get more than 7,300 MB/s in simple Sequential Read benchmarks which is quite impressive considering that PCIe Gen 4.0 x4 bandwidth is 8 Gbps and most of the competition only comes close to hitting 7,000 MB/s. What's even better is that it has higher Sequential Write speeds of 6,100 MB/s whereas most of the other competitors only come close to 5,500 MB/s. So on both fronts, the Seagate FireCuda 530 performs better. Then there's the latency aspect which decides the responsiveness of the SSD and the FireCuda 530 scored 49 ms in PCMark storage benchmark and in our benchmark to test game level loading times, the FireCuda was the fastest to load up in under 7.5 seconds. It's interesting to see how the same controller can be leveraged in so many different ways across different SKUs, and Seagate got it working at its hardest with the FireCuda 530 and that's why it gets this year's Zero1 Award for NVMe SSDs.



Seagate FireCuda 530 NVMe SSD
Price: ₹18,999

BEST BUY digit DECEMBER 2021

WD Black SN850 NVMe SSD
Price: ₹15,900

AORUS Gen4 7000s
Price: ₹18,499

1st Runner Up & Best Buy: WD Black SN850 NVMe SSD

It's been nearly a year since the WD Black SN850 was launched and it's still one of the fastest SSDs in the market. Based on a proprietary SanDisk microcontroller, the SN850 was the company's first PCIe Gen 4.0 SSD and uses their own 96-Layer BiCS4 3D NAND. Western Digital and Kioxia have since then come out with the faster 112-Layer BiCS5 3D NAND but those are yet to pop-up in any of the consumer grade SSDs. Even without their latest tech, the WD Black SN850 is a force to reckon with since it nears the 7,100 MB/s mark in Sequential Read benchmarks and gets about 5,300 MB/s on the Sequential Write benchmarks. Latency-wise, it's not that bad either from this year's winner. And in our game level load time benchmark, the SN850 finished in a little over 10 seconds. If being the First Runner-up wasn't enough, the WD Black SN850 is so economical that it even gets the Best Buy mention this year.

2nd Runner Up: AORUS Gen4 7000s

Featuring the same Phison E18 controller as this year's winner, the GIGABYTE AORUS Gen4 7000s differs by using Micron's 96-Layer TLC 3D NAND. Performance-wise, the Gen4 7000s scores about the same 7,100 MB/s mark in Sequential Read benchmarks as the WD Black SN850 and even hits 5,300 MB/s in the Sequential Write benchmarks but the one major factor that made it lag behind the competition was the IOPS figure of 350K versus the 920K of the WD Black SN850 and 700K of the Seagate FireCuda 530. When it comes to gaming, the Gen4 7000s completed the level load benchmark in under 8.5 seconds. All these factors put together make the AORUS Gen4 7000s the Second Runner-Up.

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